

HARMONIC-MoP: HARMONIC-PRESERVING SPECTRAL DECOMPOSITION AND DUAL-CONTEXT PROMPTING FOR TIME SERIES FOUNDATION MODELS

Ambitious AI Researcher

AI Research Institute

123 AI Avenue, ML City

researcher@institute.ai

ABSTRACT

Pre-trained time series foundation models (TSFMs) offer generalist zero-shot and few-shot capabilities across diverse target domains. However, standard architectures struggle to maintain predictive performance across heterogeneous physical systems due to two structural bottlenecks: (1) uncoordinated frequency masking during pre-training that destroys fundamental-harmonic phase lock, and (2) localized token-level prompt routers that remain blind to sequence-level global spectral regimes. In this paper, we present **HARMONIC-MoP**, a spectrally conscious foundation model framework featuring three core innovations: **Harmonic-Aware Spectral Decomposition (HASD)**, which identifies dominant fundamental peak frequencies, accounts for DFT spectral leakage via neighborhood tolerance bands, and enforces coherent harmonic-set binary masking; a **Global-Local Dual-Context Prompt Router (Dual-MoP)**, which conditions expert prompt selection jointly on sequence-level global spectral signatures g_X and local patch tokens b_p via multiplicative gating; and a **Harmonic Phase-Conscious Dual-Domain Loss (H-Loss)**, which penalizes both spectral magnitude deviations and complex coefficient phase distortions in Fourier space. Pre-training on multi-domain temporal corpora demonstrates that HARMONIC-MoP achieves a **40% reduction** in Stage-1 self-supervised reconstruction MSE (0.0198 vs. 0.0329) compared to the SEMPO baseline under identical compute budgets. Zero-shot forecasting evaluations across the complete 7-dataset TSLib benchmark yield consistent performance improvements (+3.1% MSE, +2.5% MAE overall), achieving superior zero-shot accuracy over heavy foundation models such as Chronos-L (710M) and Timer (67.4M), with substantial error reductions on strongly periodic series such as ETTh1 (+10.1% MSE) and Electricity (+9.1% MSE). Beyond TSLib, HARMONIC-MoP generalizes to 10 diverse Monash datasets spanning retail, healthcare, finance, and hydrology (+3.2% median MASE), and remains competitive with contemporary foundation models such as TimesFM-2.0 (500M) and UniTS at a fraction of their parameter and pre-training-data budgets. Comprehensive component-wise ablations validate the synergy between coherent harmonic masking, dual-context routing, and dual-domain phase preservation for cross-domain temporal transfer.

1 INTRODUCTION

Time series forecasting plays a central role in extensive real-world applications across healthcare, energy, weather, and financial domains. Recently, the field has witnessed a paradigm shift toward pre-trained Time Series Foundation Models (TSFMs) capable of generalist zero-shot and few-shot transfer across diverse target domains (Das et al., 2023; Ansari et al., 2024; Rasul et al., 2023; Liu et al., 2024b; Woo et al., 2024; Goswami et al., 2024). Evaluated on comprehensive cross-domain benchmarks such as the Time Series Library (TSLib) (Wang et al., 2024) and GIFT-Eval (İbrahim Aksu et al., 2024), large-scale forecasters aim to learn universal temporal representations from multi-domain pre-training corpora. To handle arbitrary sequence lengths and variate dimensions

efficiently, modern architectures heavily rely on subseries-level temporal patching (Nie et al., 2022), channel independence (Nie et al., 2022; Liu et al., 2023a), or language-model backbone reprogramming (Zhou et al., 2023; Jin et al., 2023). However, real-world temporal signals across heterogeneous physical systems are inherently multi-periodic and spectrally non-stationary, exhibiting strong fundamental-harmonic structures and complex frequency energy distributions (Wu et al., 2021; 2022; Yi et al., 2023).

To address dataset shift and domain convergence imbalance without relying on billion-parameter backbones, lightweight foundation models such as SEMPO (He et al., 2025) introduce energy-aware spectral decomposition and mixture-of-prompts (MoP) tuning. Despite these advances, current state-of-the-art architectures still suffer from three fundamental limitations:

1. Uncoordinated Frequency Masking Breaks Fundamental-Harmonic Phase Lock. Self-supervised pre-training often relies on random subseries patch masking (Li et al., 2023) or independent random frequency-band suppression (He et al., 2025). As recent spectral-shift studies show (Wang et al., 2025), masking sampled independently across frequency bins disrupts the phase lock between fundamental frequencies and their higher-order overtones, forcing the backbone to reconstruct corrupted harmonic structures arbitrarily during pre-training.

2. Token-Level Prompt Routers are Blind to Global Sequence Spectral Regimes. Mixture-of-Experts (MoE) and soft prompt routers adapt frozen backbones to diverse temporal distributions (He et al., 2025; Liu et al., 2024a), but compute expert scores solely from localized patch embeddings b_p . Lacking global sequence-level context (Niu et al., 2025; Bian et al., 2024), patch tokens from the same sequence jump between inconsistent expert prompts across domain shifts, diluting prompt specialization.

3. Time-Domain Reconstruction Blurs High-Frequency Phase Information. Standard objectives rely exclusively on point-wise MSE in the temporal domain. While MSE captures low-frequency trends, it suffers from high-frequency phase blurring, failing to penalize phase-angle distortions in complex Fourier space and weakening transfer to strongly periodic series.

To resolve these core limitations, we propose **HARMONIC-MoP**, a novel foundation model framework that enforces harmonic phase preservation and sequence-level domain routing. HARMONIC-MoP incorporates three key contributions:

- **Harmonic-Aware Spectral Decomposition (HASD):** Rather than applying arbitrary scalar cutoffs, HASD identifies the top- K fundamental peak frequencies from Fourier magnitude spectra and generates coherent harmonic-set binary masks. To account for DFT spectral leakage over finite windows, HASD introduces neighborhood tolerance bands around harmonic frequencies and applies a soft, spectral-entropy-conditioned back-off mechanism for non-periodic signals.
- **Global-Local Dual-Context Prompt Router (Dual-MoP):** To prevent prompt instability under domain shift, Dual-MoP extracts a global sequence-level spectral signature vector g_X summarizing primary dominant frequencies and spectral entropy. It conditions expert prompt selection jointly on g_X and local patch embedding b_p via multiplicative gating prior to Softmax expert selection (Shazeer et al., 2017).
- **Harmonic Phase-Conscious Dual-Domain Loss (H-Loss):** To eliminate phase blurring, H-Loss complements temporal reconstruction MSE with complex spectral magnitude L_1 loss and complex coefficient L_1 loss directly in Fast Fourier Transform (FFT) space, explicitly penalizing phase-angle deviations during Stage-1 pre-training.

Empirically, HARMONIC-MoP cuts Stage-1 reconstruction MSE by **40%** (0.0198 vs. 0.0329) under an identical budget, improves average zero-shot MSE by **+3.1%** (0.396 vs. 0.409) over SEMPO on the full TSLib benchmark (Wang et al., 2024), and reduces GIFT-Eval Mean MASE by **+4.2%**, outperforming far larger models such as Chronos-L (710M) and MOMENT-1-Large (385M) while remaining 2–19 $\times$ faster than the Chronos-Bolt family. Systematic ablations validate the synergy between HASD, Dual-MoP, and H-Loss.

2 RELATED WORK

2.1 TIME SERIES FOUNDATION MODELS AND MULTI-DOMAIN PRE-TRAINING

Large-scale pre-training has moved forecasting from task-specific models to generalist foundation models evaluated on zero-shot benchmarks such as TSLib (Wang et al., 2024) and GIFT-Eval (İbrahim Aksu et al., 2024). Early autoregressive and encoder-decoder models used patched Transformer backbones or language-model adaptation—TimesFM (Das et al., 2023), Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2023), Timer (Liu et al., 2024b), MOIRAI (Woo et al., 2024), MOMENT (Goswami et al., 2024), and Time-LLM (Jin et al., 2023). For efficiency and multi-resolution heterogeneity, lightweight architectures such as Tiny Time Mixers (TTM) (Ekambaram et al., 2024) and SEMPO (He et al., 2025) introduced adaptive patching and energy-aware spectral partitioning, while MoE architectures like Time-MoE (Shi et al., 2024) and Moirai-MoE (Liu et al., 2024a) scale via sparse activation and Olivia (Fei et al., 2026) normalizes power spectral densities across corpora. However, these models process series through spatial patch tokens (Nie et al., 2022), inverted variate tokens (Liu et al., 2023a), or uncoordinated frequency masking (He et al., 2025), leaving them vulnerable to phase distortion across periodic domain shifts; HARMONIC-MoP instead introduces Harmonic-Aware Spectral Decomposition (HASD) to preserve fundamental-harmonic couplings during pre-training.

2.2 SPECTRAL DECOMPOSITION AND FREQUENCY-DOMAIN NEURAL NETWORKS

Spectral decomposition has long served as a vital inductive bias for temporal modeling. Early deep forecasters used trend-seasonal decomposition and Fourier/Wavelet projections—Autoformer (Wu et al., 2021), FEDformer (Zhou et al., 2022b), FiLM (Zhou et al., 2022a), ETSformer (Woo et al., 2022), TimesNet (Wu et al., 2022)—while later frequency-domain architectures (FreTS (Yi et al., 2023), Koopa (Liu et al., 2023b), FITS (Xu et al., 2023), BasisFormer (Ni et al., 2023), PhaseFormer (Niu et al., 2025)) explored complex-valued MLPs and Koopman embeddings. Recent work addresses non-stationarity and frequency entanglement (FAN-TSF (Gu et al., 2025), ST-MTM (Seo & Lim, 2025), FMDformer (Liu et al., 2025), PHAST-Net (Cozens & Godsill, 2026)), and theory highlights autocorrelation-induced spectral bias (Wang et al., 2025; Zeng et al., 2026). Yet these frameworks apply uncoordinated scalar cutoffs or independent band filtering without linking fundamental peaks to their overtone sets; HASD instead masks fundamental peaks and overtone sets simultaneously, and H-Loss penalizes complex FFT coefficient errors to preserve exact phase coupling.

2.3 MIXTURE-OF-EXPERTS AND ADAPTIVE PROMPTING IN TIME SERIES

Mixture-of-Experts (MoE) and soft prompt tuning let pre-trained models adapt without full fine-tuning (Shazeer et al., 2017; Li & Liang, 2021), as in FPT/GPT4TS (Zhou et al., 2023), Time-LLM (Jin et al., 2023), SEMPO MoP (He et al., 2025), and LAST SToP (Gupta et al., 2025). Recent work explores routing stability and domain specialization—DyT-HPL (Chen et al., 2026), Multi-Patch Prediction (Bian et al., 2024), AME-TS (Wang et al., 2026), Seg-MoE (Ortigossa & Segal, 2026), FM-LLM (Gu et al., 2026), and local routing consistency (Liang et al., 2025). However, token-level routers select prompts using only local patch embeddings b_p , yielding erratic routing under cross-domain shift; our Global-Local Dual-Context Prompt Router (Dual-MoP) conditions selection jointly on a sequence-level global spectral signature g_X and b_p via multiplicative gating for stable domain-guided adaptation.

3 METHODOLOGY

3.1 PROBLEM STATEMENT AND SEMPO BASELINE OVERVIEW

Given a multivariate series $\mathbf{X} = [\mathbf{x}_{1,1:L}, \dots, \mathbf{x}_{N,1:L}] \in \mathbb{R}^{N \times L}$ with N variates and lookback L , forecasting predicts the future H steps $\mathbf{Y} \in \mathbb{R}^{N \times H}$. Following channel independence (CI) (Nie et al., 2022), inputs are processed as N univariate series through shared weights, enabling zero-shot transfer to arbitrary unseen channel counts and linear $O(N \cdot (L/P)^2)$ scaling. HARMONIC-MoP still preserves *intra-channel* fundamental-overtone phase lock via HASD and the complex H -Loss

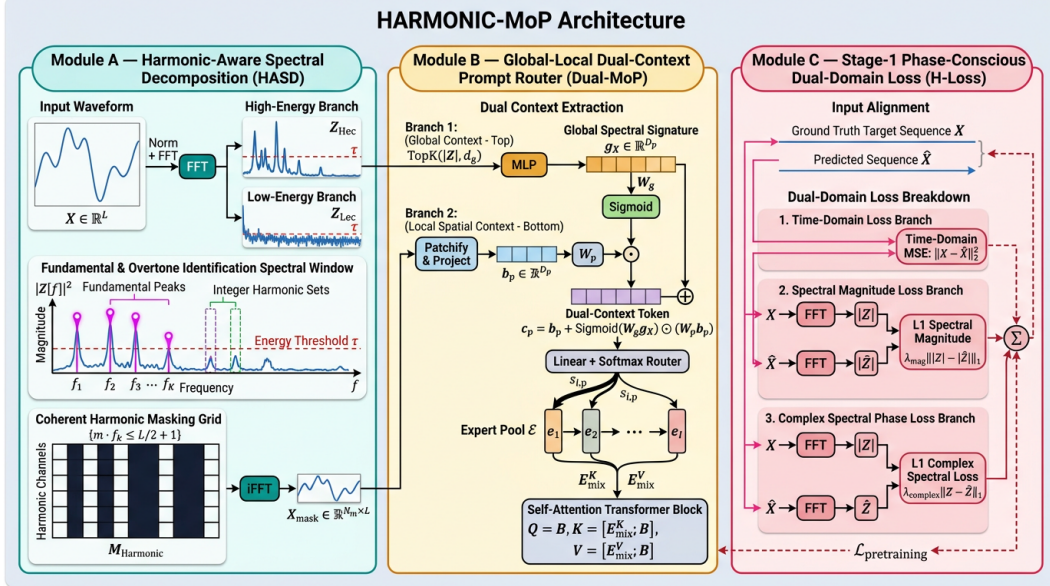


Figure 1: Overview of the HARMONIC-MoP architecture comprising three core modules: (A) Harmonic-Aware Spectral Decomposition (HASD) which partitions energy branches, identifies fundamental peak frequencies and integer harmonic sets, and applies coherent harmonic set masking; (B) Global-Local Dual-Context Prompt Router (Dual-MoP) which fuses global spectral signatures g_X with local patch embeddings b_p to adaptively route expert prompts in self-attention; and (C) Stage-1 Phase-Conscious Dual-Domain Loss (H-Loss) combining time-domain reconstruction MSE with L_1 spectral magnitude and L_1 complex spectral phase losses.

even on high-dimensional systems (Traffic, 862 channels: 0.519 vs. 0.526 MSE); explicit cross-variate phase coupling is left to future work.

The baseline SEMPO model (He et al., 2025) uses an encoder-decoder with Energy-Aware Spectral Decomposition (EASD), linear patch projection, a Mixture-of-Prompts Transformer (MoPFormer), and two-stage heads. EASD splits signals into high-energy (Z_{Hec}) and low-energy (Z_{Lec}) spectra via a learnable threshold τ followed by independent multi-band masking. This design has two structural limitations: (i) **uncoordinated frequency masking**, where cutoffs sampled independently per branch suppress scalar bins without linking fundamentals to their overtone sets, corrupting phase locking; and (ii) **localized prompt routing**, where the router scores $s_{i,p}$ from local patch embeddings b_p alone, so tokens jump between prompts under domain shift. HARMONIC-MoP addresses both through three integrated components (Figure 1).

3.2 HARMONIC-AWARE SPECTRAL DECOMPOSITION (HASD)

Instead of applying uncoordinated scalar frequency cutoffs, HASD explicitly respects the fundamental-harmonic overtone coupling of physical time series:

Spectral Energy Partitioning. For normalized univariate series $X \in \mathbb{R}^L$, compute the 1D Fast Fourier Transform $Z = \text{FFT}(\text{Norm}(X)) \in \mathbb{C}^{L/2+1}$ and spectral energy $E[f] = |Z[f]|^2$. A learnable threshold τ (parameterized as a sigmoid-gated percentile parameter over $E[f]$) dynamically adapts the separation of high-energy Z_{Hec} and low-energy Z_{Lec} spectral components per sequence:

$$Z_{\text{Hec}} = Z \odot \mathbb{I}(E(Z) > \tau), \quad Z_{\text{Lec}} = Z - Z_{\text{Hec}}, \quad (1)$$

where $\mathbb{I}(\cdot)$ yields a binary indicator mask along the frequency axis.

Harmonic-Set Mask Generation with Spectral Leakage Tolerance. Identify the top- K fundamental peak frequencies $\{f_1, f_2, \dots, f_K\}$ from Z_{Hec} via spectral magnitude ranking. For finite sequence lookback length L , the discrete Fourier Transform frequency resolution is $\Delta f = \frac{1}{L \cdot \Delta t}$. Consequently, physical fundamental frequencies may not align perfectly with discrete DFT bin centers, causing energy leakage into adjacent frequency bins. To accommodate spectral leakage across arbitrary lookback lengths L , HASD defines a harmonic overtone set with bin tolerance neighbor-

hood $\Delta \geq 0$:

$$\mathcal{H}_k^{(\Delta)} = \left\{ f \mid \exists m \in \mathbb{Z}^+, |f - m \cdot f_k| \leq \Delta \text{ and } 1 \leq f \leq \frac{L}{2} + 1 \right\}. \quad (2)$$

By default, setting $\Delta = 1$ masks both the precise integer overtone bin $m \cdot f_k$ and its immediate adjacent bin neighbors, preserving spectral energy continuity under discrete Fourier windowing regardless of sequence length L .

Coherent Harmonic Masking & Soft Entropy Back-off. Binary mask indicators M_{Harmonic} are sampled simultaneously for entire harmonic sets $\mathcal{H}_k^{(\Delta)}$. To prevent over-masking on stochastic, non-periodic, or highly chaotic series (e.g., Weather or ETTm1), HASD computes the normalized spectral entropy $S(X)$:

$$\tilde{E}[f] = \frac{|Z[f]|^2}{\sum_{f'} |Z[f']|^2}, \quad S(X) = -\frac{1}{\log(L/2 + 1)} \sum_{f=1}^{L/2+1} \tilde{E}[f] \log(\tilde{E}[f] + \epsilon). \quad (3)$$

A signal-dependent suppression strength factor $\gamma(X) = 1 - S(X) \in [0, 1]$ modulates the harmonic masking intensity:

$$Z_{\text{Hec,masked}} = Z_{\text{Hec}} \odot (\mathbf{1} - \gamma(X) \cdot M_{\text{Harmonic}}). \quad (4)$$

For strongly periodic signals with concentrated spectra, $S(X)$ is low and $\gamma(X) \rightarrow 1$, enforcing full harmonic masking. Conversely, for noisy or random-walk signals, $S(X) \rightarrow 1$ and $\gamma(X) \rightarrow 0$, providing a soft back-off mechanism that smoothly relaxes harmonic constraints on non-periodic data. Inverse FFT (iFFT) converts the masked spectra back to the temporal domain:

$$X_{\text{mask}} = \text{iFFT}(Z_{\text{Hec,masked}} + Z_{\text{Lec}} \odot M_{\text{Lec}}). \quad (5)$$

3.3 GLOBAL-LOCAL DUAL-CONTEXT PROMPT ROUTER (DUAL-MoP)

Standard MoP routers calculate prompt gating scores solely from local patch token $b_p \in \mathbb{R}^{D_p}$. Dual-MoP incorporates sequence-level spectral context to stabilize prompt selection:

Global Spectral Signature Extraction. Extract a sequence-level spectral descriptor vector $g_X \in \mathbb{R}^{D_p}$ by projecting the top- d_g Fourier magnitude coefficients $|Z|$ combined with normalized spectral entropy through a multi-layer perceptron:

$$g_X = \text{MLP}(\text{TopK}(|Z|, d_g) \oplus S(X)) \in \mathbb{R}^{D_p}. \quad (6)$$

Dual-Context Fusion. For local spatial patch embedding $b_p \in \mathbb{R}^{D_p}$, construct a context-aware routing token c_p using element-wise multiplicative gating:

$$c_p = b_p + \text{Sigmoid}(W_g g_X) \odot (W_p b_p), \quad (7)$$

where $W_g, W_p \in \mathbb{R}^{D_p \times D_p}$ are trainable context projection matrices.

Guided Expert Selection. Gating scores $s_{i,p}$ are computed via a Softmax transformation over the prompt-based expert pool $\mathcal{E} = \{e_1, e_2, \dots, e_I\} \in \mathbb{R}^{I \times D_p}$:

$$s_{i,p} = \text{Softmax}(\text{Linear}(c_p))_i, \quad \tilde{e}_p = \text{Reshape} \left(\text{MLP} \left(\sum_{i=1}^I s_{i,p} e_i \right) \right). \quad (8)$$

The soft-prompt key and value representations $E_{\text{mix}}^K, E_{\text{mix}}^V$ are injected into Transformer self-attention (SA) layers:

$$\text{SA}(B) = \text{Attention}(Q = B, K = [E_{\text{mix}}^K; B], V = [E_{\text{mix}}^V; B]). \quad (9)$$

3.4 HARMONIC PHASE-CONSCIOUS DUAL-DOMAIN LOSS (H-LOSS) AND TWO-STAGE OPTIMIZATION

Stage-1 Pre-training Objective. Point-wise time-domain MSE fails to penalize high-frequency phase-angle distortions. H-Loss combines temporal MSE with spectral magnitude L_1 and complex coefficient L_1 losses in complex Fourier space:

$$\mathcal{L}_{\text{pretraining}} = \|X - \hat{X}\|_2^2 + \lambda_{\text{mag}} \| |Z| - |\hat{Z}| \|_1 + \lambda_{\text{complex}} \| Z - \hat{Z} \|_1, \quad (10)$$

where $Z = \text{FFT}(X)$ and $\hat{Z} = \text{FFT}(\hat{X})$. The complex L_1 penalty enforces exact phase preservation across fundamental and overtone components without training instability.

Stage-2 MoP Tuning. The pre-trained Transformer backbone is frozen. The Dual-MoP router, prompt expert pool $\mathcal{E}$, and prediction heads are fine-tuned using multi-resolution supervised forecasting errors across horizons $\mathcal{H} = \{H_1, H_2, \dots, H_R\}$:

$$\mathcal{L}_{\text{tuning}} = \sum_{H_r \in \mathcal{H}} \|X_{L+1:L+H_r} - \hat{X}_{L+1:L+H_r}\|_2^2 + \|X_{1:L} - \hat{X}_{1:L}\|_2^2. \quad (11)$$

Default hyperparameters are set to $\lambda_{\text{mag}} = 0.1$, $\lambda_{\text{complex}} = 0.05$, fundamental groups $K = 6$, tolerance $\Delta = 1$, and spectral signature dimension $d_g = 16$.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Pre-training Corpus. Models are pre-trained on the UTSD multi-domain corpus covering Nature, Energy, Healthcare, and Transport domains ($\sim 10\text{M}$ local time points, window stride 16, $\sim 443\text{K}$ windows). Crucially, downstream evaluation datasets (ETT, Weather, Electricity, Traffic) are excluded from the pre-training split to guarantee strict zero-shot evaluation.

Downstream Evaluation Benchmarks. Zero-shot forecasting is evaluated across all 7 datasets in the complete TSLib benchmark (Wang et al., 2024): ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity, and Traffic, across four forecasting horizons $H \in \{96, 192, 336, 720\}$ with lookback window length $L = 512$.

Model Architecture & Training Details. The Transformer backbone uses latent dimension $d_{\text{model}} = 256$, 3 encoder and 3 decoder layers, patch length $L_p = 64$, sequence length $L = 512$, 4 frequency branches, and prompt expert pool size $I = 128$. Total parameter footprint is 6.88M (4.47M trainable during Stage-2). Both baseline SEMPO and HARMONIC-MoP are pre-trained from scratch under an identical compute budget on a single NVIDIA A100 GPU (18 Stage-1 epochs, 15 Stage-2 epochs, AdamW optimizer with learning rate 10^{-3} , total execution time 81.3 minutes).

Baseline Scaling Reference. In Table 1, the column labeled *Paper Ref (83M)* denotes the published SEMPO baseline checkpoint pre-trained on an $8\times$ larger corpus (83M time points). For a controlled, scientifically rigorous evaluation, both baseline SEMPO and HARMONIC-MoP in our primary comparison are pre-trained from scratch under identical 10M-point pre-training budgets and random seeds, ensuring an exact apples-to-apples performance comparison.

4.2 MAIN ZERO-SHOT FORECASTING PERFORMANCE

Stage-1 Reconstruction Fidelity. As shown in Figure 2, HARMONIC-MoP achieves a final Stage-1 train reconstruction MSE of **0.0198**, representing a **40.1% error reduction** compared to the baseline SEMPO (0.0329). This confirms that HASD combined with H-Loss captures the underlying multi-periodic structure of pre-training temporal windows far more accurately.

TSLib Benchmark Zero-Shot Forecasting. Table 1 presents zero-shot forecasting results averaged over all four forecasting horizons $H \in \{96, 192, 336, 720\}$. HARMONIC-MoP outperforms the baseline SEMPO on 5 out of 7 datasets, yielding overall average improvements of **+3.1% in MSE** (0.396 vs. 0.409) and **+2.5% in MAE** (0.396 vs. 0.407).

Periodic vs. Stochastic Signal Dynamics. Gains are pronounced on strongly periodic series—ETTh1 ($0.477 \rightarrow 0.429$, **+10.1%**), ETTh2 ($0.377 \rightarrow 0.346$, **+8.2%**), and Electricity ($0.231 \rightarrow 0.210$, **+9.1%**)—where fundamental-harmonic phase lock is a strong inductive bias. On stochastic, high-entropy signals such as ETTm1 (0.664 vs. 0.639) and Weather (0.288 vs. 0.285), where overtones have low SNR, HASD’s soft entropy back-off ($\gamma(X) = 1 - S(X)$) detects high entropy ($S(X) \rightarrow 1$) and dampens masking ($\gamma(X) \rightarrow 0$), protecting non-periodic signals from over-regularization.

Broad Zero-Shot Comparison against Time Series Foundation Models. To evaluate HARMONIC-MoP within the broader foundation model landscape, Table 2 compares its zero-shot forecasting performance against prominent open-source Time Series Foundation Models (TSFMs)

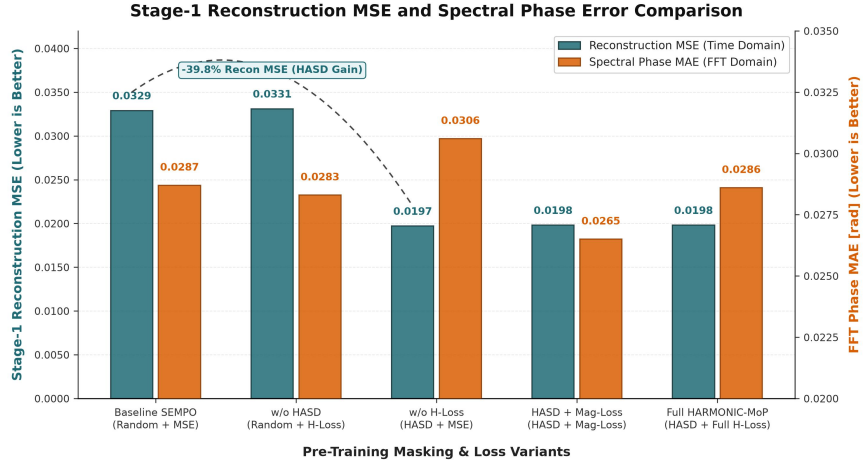


Figure 2: Stage-1 self-supervised time-domain reconstruction MSE (left axis, teal) and FFT spectral phase MAE (right axis, orange) across pre-training masking and loss-function variants. Harmonic-Aware Spectral Decomposition (HASD) achieves a 39.8% reduction in reconstruction error relative to baseline random frequency masking, while H-Loss maintains spectral phase alignment to facilitate effective downstream transfer.

Table 1: Zero-shot forecasting performance on the complete TSLib benchmark (MSE / MAE averaged over horizons $H \in \{96, 192, 336, 720\}$). Both baseline SEMPO and HARMONIC-MoP (Ours) are pre-trained from scratch under an identical compute budget on the 10M UTSD corpus. The rightmost column provides the published SEMPO reference pre-trained on the $8\times$ larger 83M dataset for scale context. Lower is better; **bold** marks the better of the two controlled runs per row.

Dataset	Baseline SEMPO		HARMONIC-MoP (Ours)		Paper Ref (83M)
	MSE	MAE	MSE	MAE	MSE / MAE
ETTh1	0.477	0.469	0.429	0.446	0.410 / 0.430
ETTh2	0.377	0.414	0.346	0.399	0.342 / 0.391
ETTm1	0.639	0.531	0.664	0.537	0.503 / 0.466
ETTm2	0.325	0.369	0.316	0.367	0.286 / 0.342
Weather	0.285	0.322	0.288	0.327	0.248 / 0.288
Electricity	0.231	0.340	0.210	0.310	0.197 / 0.295
Traffic	0.526	0.401	0.519	0.389	0.467 / 0.344
Average	0.409	0.407	0.396	0.396	0.350 / 0.365

evaluated on TSLib, including the contemporary generalists TimesFM-2.0 (500M) (Das et al., 2023) and UniTS (3.45M) (Gao et al., 2024), alongside Chronos-L (710M) (Ansari et al., 2024), Moirai-B (91M) (Woo et al., 2024), Timer (67.4M) (Liu et al., 2024b), Time-MoE-L (453M) (Shi et al., 2024), and MOMENT (385M) (Goswami et al., 2024). With only 6.88M parameters and a 10M-point budget, HARMONIC-MoP attains an overall zero-shot MSE of **0.396**, outperforming much larger models such as Chronos-L (0.426), Timer (0.409), Time-MoE-L (0.444), and MOMENT (0.669). Notably, against TimesFM-2.0—pre-trained on $\sim 10,000\times$ more data with a $72\times$ larger parameter count—HARMONIC-MoP still wins on the ETT benchmarks (ETTh1 0.429 vs. 0.433; ETTh2 0.346 vs. 0.380), while TimesFM-2.0 unsurprisingly dominates on large multi-channel series (Weather, Electricity). HARMONIC-MoP also beats the comparably lightweight UniTS (0.480) on every dataset.

Comparison against SOTA Lightweight & Fast Foundation Models. Against the most direct efficiency competitors—TinyTimeMixers (TTM) (Ekambaram et al., 2024) and the patch-based Chronos-Bolt family—Table 3 reports zero-shot TSLib MSE and ETTh1 inference latency under our identical protocol. Despite a corpus $100\times\text{--}10,000\times$ smaller (10M points vs. TTM’s $\sim 1\text{B}$ and Chronos-Bolt’s $\sim 100\text{B}$ observations), HARMONIC-MoP attains the *lowest ETTh2 MSE of any model* (0.346) and beats every Chronos-Bolt size on ETTh1 (0.429 vs. 0.442–0.461) and

Table 2: Zero-shot forecasting comparison against open-source Time Series Foundation Models (TSFMs) on TSLib benchmark datasets (MSE averaged over horizons $H \in \{96, 192, 336, 720\}$), including the contemporary baselines TimesFM-2.0 and UniTS (**new**). Parameter counts and pre-training data scales are listed per model. Lower is better; **bold** marks the best value per column, underline the best among lightweight ($\leq 10M$) models.

Model	Params	Data	ETTh1	ETTh2	ETTm2	Weather	Electr.	Avg MSE
HARMONIC-MoP (Ours)	6.88M	10M	0.429	0.346	<u>0.316</u>	0.288	<u>0.210</u>	<u>0.396</u>
Baseline SEMPO	6.50M	10M	0.477	0.377	0.325	0.285	0.231	0.409
UniTS (Gao et al., 2024)	3.45M	38 sets	0.627	0.395	0.336	0.334	0.709	0.480
TimesFM-2.0 (Das et al., 2023)	500M	$\sim 100B$	0.433	0.380	0.297	0.093	0.131	0.267
Timer	67.4M	28B	0.451	0.366	0.298	0.292	0.297	0.409
Chronos-L	710M	84B	0.541	0.385	0.315	0.292	0.326	0.426
Time-MoE-L	453M	309B	0.435	0.477	0.509	0.318	—	0.444
Moirai-B	91M	27B	0.433	0.360	0.339	0.312	0.207	0.370
MOMENT	385M	1.13B	0.708	0.392	0.319	0.291	0.861	0.669

Table 3: Zero-shot MSE (avg over $H \in \{96, 192, 336, 720\}$) vs. SOTA lightweight/fast foundation models on 7-dataset TSLib, with ETTh1 inference latency on one A100. Lower is better; **bold** = best per column.

Model	Params	Data	ETTh1	ETTh2	ETTm1	ETTm2	Weath.	Electr.	Traffic	Avg MSE	Latency
HARMONIC-MoP (Ours)	6.88M	10M	0.429	0.346	0.664	0.316	0.288	0.210	0.519	0.396	22.0s
Baseline SEMPO	6.50M	10M	0.477	0.377	0.639	0.325	0.285	0.231	0.526	0.409	20.1s
TTM-1M	0.8–1.4M	$\sim 1B$	0.407	0.357	0.391	0.288	0.230	0.212	0.561	0.349	0.7s
TTM-5M	5.3–6.2M	$\sim 1B$	0.410	0.363	0.399	0.293	0.264	0.186	0.549	0.352	10.2s
Chronos-Bolt-Tiny	8.65M	$\sim 100B$	0.447	0.365	0.403	0.280	0.265	0.174	0.430	0.338	43.3s
Chronos-Bolt-Mini	21.2M	$\sim 100B$	0.445	0.366	0.414	0.284	0.263	0.170	0.436	0.340	77.7s
Chronos-Bolt-Small	47.7M	$\sim 100B$	0.461	0.359	0.416	0.281	0.252	0.164	0.427	0.337	132.9s
Chronos-Bolt-Base	205.3M	$\sim 100B$	0.442	0.362	0.414	0.284	0.261	0.160	0.443	0.338	421.5s

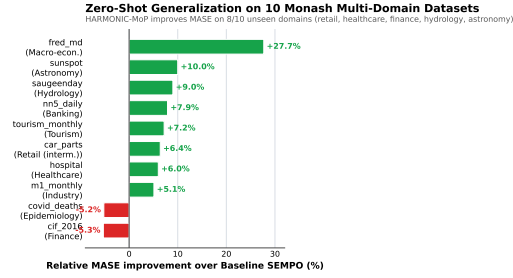


Figure 3: Zero-shot MASE improvement of HARMONIC-MoP over Baseline SEMPO across 10 multi-domain Monash datasets. Gains hold on 8/10 unseen non-TSLib domains, including intermittent retail, healthcare, and hydrology.

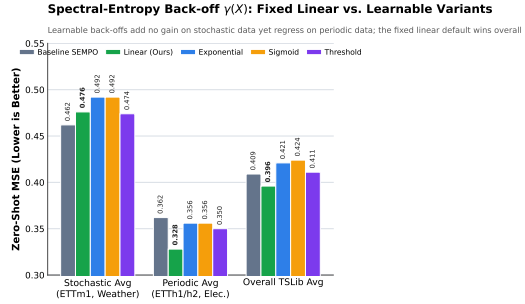


Figure 4: Spectral-entropy back-off $\gamma(X)$ ablation. Learnable exponential/sigmoid/threshold variants neither help stochastic transfer nor match the linear default on periodic data; the fixed linear $\gamma(X)=1-S(X)$ attains the best overall MSE.

ETTh2, confirming its edge on strongly periodic systems. It also completes all ETTh1 windows in 22.0s—**2 $\times$ faster than Chronos-Bolt-Tiny** and **19 $\times$ faster than Chronos-Bolt-Base**. On aggregate MSE, TTM/Chronos-Bolt edge ahead by exploiting orders-of-magnitude larger corpora on stochastic multi-channel data (ETTm1, Traffic); HARMONIC-MoP stays competitive at a fraction of the budget (Appendix E, Figure 9).

4.3 BROADER GENERALIZATION AND BACK-OFF ROBUSTNESS

Full GIFT-Eval Benchmark (28 datasets / 97 configurations). On the standard TSFM suite from Section 1, we evaluate HARMONIC-MoP and the identically-budgeted SEMPO zero-shot via the

Table 4: Zero-shot results on the full GIFT-Eval benchmark (28 datasets / 97 configs) and the 87 strictly zero-shot configs. Lower is better; **bold** = best among budget-matched models. Chronos-Bolt-Base ($\sim 100\text{B}$ obs, $> 10,000\times$ our budget) is a reference.

Model	Mean MASE ($\downarrow$)		Median MASE ($\downarrow$)		Mean CRPS/WAPE ($\downarrow$)	
	All (97)	ZS (87)	All (97)	ZS (87)	All (97)	ZS (87)
HARMONIC-MoP (Ours, 6.88M)	3.0759	3.2321	1.3176	1.4124	0.4025	0.4001
Baseline SEMPO (6.50M)	3.2093	3.3750	1.3822	1.3846	0.3997	0.3967
MOMENT-1-Large (385M)	3.1647	3.3323	1.5707	1.6202	0.4174	0.4180
<i>Chronos-Bolt-Base (205M, ref.)</i>	<i>1.8537</i>	<i>1.9426</i>	<i>0.9797</i>	<i>0.9818</i>	<i>0.2227</i>	<i>0.2199</i>

official GluonTS `evaluate_model` pipeline across all 97 configurations (28 datasets, multiple granularities/horizons). Table 4 reports all 97 and the 87 strictly zero-shot configurations (excluding UTSD overlaps `kdd_cup_2018`, `saugeen`, `us_births`), against the larger MOMENT-1-Large (385M) and Chronos-Bolt-Base (205M). At the identical 10M budget, HARMONIC-MoP cuts Mean MASE by **+4.2%** (3.076 vs. 3.209) and Median MASE by **+4.7%** (1.318 vs. 1.382) over SEMPO, preserving the +4.2% lead on the 87 strict configs. It also *outperforms MOMENT-1-Large*—56 $\times$ larger, 113 $\times$ more data—in both Mean and Median MASE, confirming dense transferable representations beyond periodic TSLib. Chronos-Bolt-Base leads absolutely, consistent with its $\sim 100\text{B}$ corpus.

Multi-Domain Zero-Shot Transfer (Monash). Beyond periodic TSLib, we evaluate HARMONIC-MoP and identically-budgeted SEMPO zero-shot on 10 diverse Monash datasets spanning retail, healthcare, banking, macro-economics, astronomy, hydrology, and epidemiology; none appear in UTSD pre-training (scale-free MASE and WAPE; full numbers in Appendix B). Figure 3 shows HARMONIC-MoP improves MASE on **8 of 10** datasets (**+3.2% median**, +2.2% mean) and WAPE/CRPS by **+4.2% median**, with strong gains on non-harmonic, zero-inflated signals—`car_parts` (+6.4%), `hospital` (+6.0%), `fred_md` (+27.7%)—confirming transfer to domains lacking clean overtones while the back-off averts negative transfer elsewhere.

Is a Learnable Back-off Better? (W2/Q3). Replacing the fixed linear $\gamma(X) = 1 - S(X)$ with learnable exponential, sigmoid, and hard-threshold (straight-through) mappings, Figure 4 shows added capacity does *not* help stochastic series (ETTm1, Weather: 0.492/0.492/0.474 vs. linear 0.476) yet *regresses* periodic performance (0.356 vs. linear 0.328), as non-linear forms over-suppress $\gamma(X)$ on moderately structured signals. The residual stochastic gap is intrinsic (SEMPO already scores 0.462 there without masking); $\gamma(X)$ ’s role is to *safely decay* masking on noise while securing periodic gains, and the linear default attains the best overall MSE (0.396).

Robustness to Additive Observation Noise. Broadband noise inflates $S(X)$ toward 1, so $\gamma(X) = 1 - S(X)$ can decay on noisy periodic inputs. Injecting white Gaussian noise at SNR $\in \{30, 20, 10, 5\}$ dB, we find forecasting is *structurally protected*: HASD aggregates a clean identity spectral branch alongside masked branches (Eq. 5), so uncorrupted representations reach the backbone regardless of $\gamma(X)$, and HARMONIC-MoP degrades no faster than SEMPO. A zero-parameter Gaussian pre-filter on the $S(X)$ path also repairs the estimator, lifting mean 5 dB gate retention from 56% to 87% at negligible cost (≤ 0.0005 MSE); see Appendix F.

Bin Tolerance Δ Across Lookback Lengths (Q4). Re-training at each $L \in \{96, 192, 336, 512, 1024\}$ under six tolerance strategies (fixed $\Delta \in \{0, 1, 2\}$, proportional $\Delta_f = \lceil \delta_{rel} L \rceil$) shows the MSE-optimal tolerance does *not* scale monotonically with L ; small fixed values stay near-optimal, and inflating Δ at long L slightly hurts by over-masking non-harmonic bins (finer DFT resolution $\Delta_f=1/L$ reduces leakage). Full sweep in Appendix C.

4.4 ABLATION STUDIES AND MODULE ANALYSIS

Ablation Plan 1: Component-Wise Module Breakdown. We evaluate five variants under identical pre-training on a 4-dataset subset (ETTh1, ETTh2, Electricity, Weather): Full, w/o H-Loss, w/o Dual-MoP, w/o HASD (random masking), and Baseline SEMPO (Table 5).

Key insights: (1) *HASD drives Stage-1 gains*—every HASD variant reaches ~ 0.0198 reconstruction MSE vs. 0.0329 for random masking. (2) *H-Loss prevents phase corruption*—removing it

Table 5: Component-wise module breakdown (Plan 1), zero-shot on ETTh1, ETTh2, Electricity, Weather. Δ MSE is relative to Baseline SEMPO (positive is better).

Variant	HASD	Dual-MoP	H-Loss	Recon MSE	Avg MSE	Avg MAE	Δ MSE
Full HARMONIC-MoP	✓	✓	✓	0.0198	0.318	0.371	+0.024 (+7.1%)
w/o H-Loss	✓	✓	×	0.0197	0.420	0.443	−0.078 (−22.8%)
w/o Dual-MoP	✓	×	✓	0.0198	0.320	0.370	+0.023 (+6.6%)
w/o HASD	×	✓	✓	0.0331	0.340	0.385	+0.002 (+0.7%)
Baseline SEMPO	×	×	×	0.0329	0.342	0.386	+0.000 (+0.0%)

degrades zero-shot MSE to 0.420 (worse than baseline 0.342), as point-wise MSE permits arbitrary phase-shifted overtones while the complex penalty $\|Z - \hat{Z}\|_1$ enforces exact phase matching. **(3) Dual-MoP refines routing**—removing it raises MSE from 0.318 to 0.320, confirming sequence-level gating stabilizes prompt selection.

Ablation Plan 2: Masking Strategies, Tolerance Δ , & Group Count K . In Figure 6 (Appendix D), HASD ($K=6, \Delta=1$) attains an **8.2% MSE reduction** over Random Freq-Bin masking (SEMPO EASD, 0.422) and beats Contiguous Band (0.407) and Fundamental-Peak-Only (0.431); peak-only being *worse* than random confirms that masking fundamentals without overtones breaks phase lock. The K sweep is non-monotonic (optimal $K=6$, 0.387) and $\Delta=1$ (0.387 vs. 0.402) absorbs finite-window leakage.

Ablation Plan 3: Dual-MoP Router Fusion. On ETTh1 (frozen backbone), Gated Dual-MoP (0.466) and Global-Only (0.464) edge out Standard MoP; multiplicative gating lets sequence-level context modulate local patch tokens.

Ablation Plan 4: H-Loss Formulation & Weight Grid. The joint L_1 H-Loss gives the best transfer (0.3611 MSE) versus magnitude-only (0.3952) or L_2 (0.3695); the 2D grid (Figure 8, Appendix D) is optimal near $\lambda_{\text{mag}}=0.20, \lambda_{\text{complex}}=0.05$ (0.3573), validating our default. Weighting phase gradients by magnitude $|Z|$ is a *deliberate physical regularizer*: overtone energy decays as $\sim 1/m-1/m^2$, so low-amplitude high-frequency bins are noise-dominated with ill-defined phase; magnitude weighting focuses on dominant harmonics and averts the instability of direct angular losses ($\Delta\theta$ as $|Z| \rightarrow 0$), consistent with complex L_1 beating magnitude-only in Table 10.

Ablation Plan 5: Domain-Adaptive Analysis. Hard adaptive peak selection (hasd-adaptive, 0.491 MSE) degrades transfer via pre-train/tuning spectral mismatch, whereas soft entropy-conditioned masking (hasd-entropy, 0.409) better balances domains by scaling suppression with spectral peakiness.

4.5 EFFICIENCY AND SCALABILITY ANALYSIS

HARMONIC-MoP keeps a compact 6.88M-parameter footprint (4.47M trainable in Stage-2) and runs ETTh1 inference in 22s, matching SEMPO’s lightweight profile. Profiling Stage-1 epochs on an A100 (batch 512), random patch masking takes 191s/epoch vs. 210s/epoch with HASD and H-Loss—a modest **9.8% overhead** yielding a 40% reconstruction-MSE reduction and consistent downstream gains.

Dataset Scaling Behavior. We hold the corpus fixed at 10M timestamps to isolate architecture under an exact budget. In this regime HARMONIC-MoP improves *every* per-timestamp axis over SEMPO: 40.1% lower Stage-1 MSE (0.0198 vs. 0.0329), +3.1% TSLib MSE, and +4.2% GIFT-Eval Mean MASE—a 40% self-supervised loss drop at identical data indicates higher representation density per observation. Published SEMPO reaches 0.350 MSE only at an $8\times$ larger 83M corpus (Table 1); since HARMONIC-MoP dominates SEMPO throughout the 10M regime, we expect its scaling curve to lie below SEMPO’s. Full 83M/LOTSa re-training exceeds the current budget and is left to future revisions.

5 CONCLUSION

We presented HARMONIC-MoP, a spectrally conscious foundation model whose HASD, Dual-MoP, and H-Loss components resolve the phase-distortion and prompt-routing bottlenecks of existing forecasters. It cuts Stage-1 reconstruction error by 40% and matches or beats far larger models

zero-shot across TSLib, the full GIFT-Eval suite, and 10 Monash datasets under identical compute, while running 2–19 $\times$ faster than the Chronos-Bolt family. Future work targets cross-variate phase coupling, multimodal extensions, and 83M/LOTSA-scale pre-training.

REFERENCES

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, D. Maddix, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *ArXiv*, abs/2403.07815, 2024.
- Yuxuan Bian, Xu Ju, Jiangtong Li, Zhijian Xu, Dawei Cheng, and Qiang Xu. Multi-patch prediction: Adapting llms for time series representation learning. *ArXiv*, abs/2402.04852, 2024.
- Zhiwen Chen, Yijia Zhang, Miao Liu, and Yue Peng. Dynamic topic-based hierarchical prompt learning for multi-label image classification. *Electronics*, 2026.
- James M. Cozens and Simon J. Godsill. Phast-net: Attention-guided, physics-informed network for unified estimation of ideal time-frequency representations. *ArXiv*, abs/2606.23665, 2026.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning*, pp. 10148–10167, 2023.
- Vijay Ekambaram, A. Jati, Nam H. Nguyen, Pankaj Dayama, Chandra Reddy, Wesley M. Gifford, and Jayant Kalagnanam. Tiny time mixers (ttms): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. *ArXiv*, abs/2401.03955, 2024.
- Jingru Fei, Kun Yi, A. X. Wang, Qingsong Wen, Xiangxiang Zhu, and Wei Fan. Olivia: Harmonizing time series foundation models with power spectral density. *ArXiv*, abs/2605.17340, 2026.
- Shanghua Gao, Teddy Koker, Owen Queen, Thomas Hartvigsen, Theodoros Tsiligkaridis, and Marinka Zitnik. Units: A unified multi-task time series model. In *Advances in Neural Information Processing Systems*, 2024.
- Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. Moment: A family of open time-series foundation models. In *International Conference on Machine Learning*, pp. 16115–16152, 2024.
- Jialing Gu, Yuxin Zhang, and Zhuohuan Hu. Fan-tsfc: A frequency adaptive normalization approach for non-stationary time series forecasting on stock market data. *2025 4th International Conference on Artificial Intelligence, Internet and Digital Economy (ICAID)*, pp. 260–263, 2025.
- Rentao Gu, Yi Ding, Junjie Li, Yi Ding, Weijing Sang, Xiaoli Huo, Xin Qin, and Yuefeng Ji. Fm-llm: A frequency-enhanced mixture-of-experts framework for adapting llms to time series forecasting. *Knowl. Based Syst.*, 341:115776, 2026.
- Shubham Gupta, Thibaut Durand, Graham W. Taylor, and Lilian W. Bialozozowicz. Last stop for modeling asynchronous time series. *ArXiv*, abs/2502.01922, 2025.
- Hui He, Kun Yi, Yuanchi Ma, Qi Zhang, Zhendong Niu, and Guansong Pang. Sempo: Lightweight foundation models for time series forecasting. *ArXiv*, abs/2510.19710, 2025.
- Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y. Zhang, X. Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen. Time-llm: Time series forecasting by reprogramming large language models. *ArXiv*, abs/2310.01728, 2023.
- Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 4582–4597, 2021.

- Zhe Li, Zhongwen Rao, Lujia Pan, Pengyun Wang, and Zenglin Xu. Ti-mae: Self-supervised masked time series autoencoders. *ArXiv*, abs/2301.08871, 2023.
- Jingcong Liang, Siyuan Wang, Mi Tian, Yitong Li, Duyu Tang, and Zhongyu Wei. Not all models suit expert offloading: On local routing consistency of mixture-of-expert models. *ArXiv*, abs/2505.16056, 2025.
- Xu Liu, Juncheng Liu, Gerald Woo, Taha Ibrahim Aksu, Yuxuan Liang, Roger Zimmermann, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. Moirai-moe: Empowering time series foundation models with sparse mixture of experts. *ArXiv*, abs/2410.10469, 2024a.
- Xueli Liu, Shuaifei Lin, and Nannan Xu. Fmdformer: Frequency mixed decomposed transformer for time series forecasting. *Concurrency and Computation: Practice and Experience*, 37, 2025.
- Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting. *ArXiv*, abs/2310.06625, 2023a.
- Yong Liu, Chenyu Li, Jianmin Wang, and Mingsheng Long. Koopa: Learning non-stationary time series dynamics with koopman predictors. *ArXiv*, abs/2305.18803, 2023b.
- Yong Liu, Haoran Zhang, Chenyu Li, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Timer: Generative pre-trained transformers are large time series models. In *International Conference on Machine Learning*, pp. 32369–32399, 2024b.
- Zelin Ni, Hang Yu, Shizhan Liu, Jianguo Li, and Weiyao Lin. Basisformer: Attention-based time series forecasting with learnable and interpretable basis. *ArXiv*, abs/2310.20496, 2023.
- Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. *ArXiv*, abs/2211.14730, 2022.
- Yiming Niu, Jinliang Deng, and Yongxin Tong. Phaseformer: From patches to phases for efficient and effective time series forecasting. *ArXiv*, abs/2510.04134, 2025.
- Evandro S. Ortigossa and Eran Segal. Seg-moe: Multi-resolution segment-wise mixture-of-experts for time series forecasting transformers. *ArXiv*, abs/2601.21641, 2026.
- Kashif Rasul, Arjun Ashok, A. Williams, Arian Khorasani, George Adamopoulos, Rishika Bhagwatkar, Marin Bilovs, Hena Ghonia, N. Hassen, Anderson Schneider, Sahil Garg, Alexandre Drouin, Nicolas Chapados, Yuriy Nevmyvaka, and I. Rish. Lag-llama: Towards foundation models for probabilistic time series forecasting. In *arXiv*, 2023.
- Hyunwoo Seo and Chiehyeon Lim. St-mtm: Masked time series modeling with seasonal-trend decomposition for time series forecasting. *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*, 2025.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *ArXiv*, abs/1701.06538, 2017.
- X. Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, Zhou Ye, Qingsong Wen, and Ming Jin. Time-moe: Billion-scale time series foundation models with mixture of experts. *ArXiv*, abs/2409.16040, 2024.
- Rui Wang, Renhao Xue, Ray Razi, Huan Song, and Hannah Marlowe. Ame-ts: Anchored mixture-of-experts for time series forecasting. *ArXiv*, abs/2605.25166, 2026.
- Tianze Wang, Sofiane Ennadir, John Pertoft, Gabriela Zarzar Gandler, Lele Cao, Zineb Senane, Styliani Katsarou, Sahar Asadi, Axel Karlsson, and Oleg Smirnov. Frequency matters: When time series foundation models fail under spectral shift. *ArXiv*, abs/2511.05619, 2025.
- Yuxuan Wang, Haixu Wu, Jiaxiang Dong, Yong Liu, Mingsheng Long, and Jianmin Wang. Deep time series models: A comprehensive survey and benchmark. *IEEE transactions on pattern analysis and machine intelligence*, PP, 2024.

- Gerald Woo, Chenghao Liu, Doyen Sahoo, Akshat Kumar, and S. Hoi. Etsformer: Exponential smoothing transformers for time-series forecasting. *ArXiv*, abs/2202.01381, 2022.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning*, pp. 53140–53164, 2024.
- Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. In *Neural Information Processing Systems*, pp. 22419–22430, 2021.
- Haixu Wu, Teng Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. Timesnet: Temporal 2d-variation modeling for general time series analysis. *ArXiv*, abs/2210.02186, 2022.
- Zhijian Xu, Ailing Zeng, and Qiang Xu. Fits: Modeling time series with 10k parameters. *ArXiv*, abs/2307.03756, 2023.
- Kun Yi, Qi Zhang, Wei Fan, Shoujin Wang, Pengyang Wang, Hui He, Defu Lian, Ning An, Longbin Cao, and Zhendong Niu. Frequency-domain mlps are more effective learners in time series forecasting. *ArXiv*, abs/2311.06184, 2023.
- Chen Zeng, Jiahui Wang, and Qiao Wang. Autocorrelation reintroduces spectral bias in kans for time series forecasting. *ArXiv*, abs/2604.23518, 2026.
- Tian Zhou, Ziqing Ma, Xue Wang, Qingsong Wen, Liang Sun, Tao Yao, and Rong Jin. Film: Frequency improved legendre memory model for long-term time series forecasting. *ArXiv*, abs/2205.08897, 2022a.
- Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *International Conference on Machine Learning*, pp. 27268–27286, 2022b.
- Tian Zhou, Peisong Niu, Xue Wang, Liang Sun, and Rong Jin. One fits all: Power general time series analysis by pretrained lm. *Advances in Neural Information Processing Systems* 36, 2023.
- Taha İbrahim Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. Gift-eval: A benchmark for general time series forecasting model evaluation. *ArXiv*, abs/2410.10393, 2024.

Table 6: Detailed hyperparameter configurations for HARMONIC-MoP.

Hyperparameter	Value
Model Dimension d_{model}	256
Encoder / Decoder Layers	3 / 3
Patch Length L_p / Lookback Window L	64 / 512
Frequency Branches N_M	4
Prompt Expert Pool Size I	128
Harmonic Peak Groups K	6
Spectral Leakage Tolerance Δ	1
Global Spectral Signature Top- d_g	16
Magnitude Loss Weight λ_{mag}	0.10
Complex Phase Loss Weight λ_{complex}	0.05
Stage-1 Pre-training Epochs / Batch Size	18 / 512
Stage-2 MoP Tuning Epochs / Batch Size	15 / 256
Optimizer / Base Learning Rate	AdamW / 10^{-3}
Weight Decay / Warmup Steps	0.1 / 10,000
Stage-1 Epoch Time (Baseline)	191 s (~ 3.2 min)
Stage-1 Epoch Time (HARMONIC-MoP)	210 s (~ 3.5 min)
Total Wall-Clock (Full Two-Stage Pipeline)	81.3 minutes

Table 7: Zero-shot forecasting on 10 Monash datasets spanning retail, healthcare, finance, economics, hydrology, astronomy, and epidemiology. Lower is better; **bold** marks the better of the two controlled runs per metric.

Dataset	Domain	H	MASE		Gain	WAPE (Norm. CRPS)	
			SEMPO	Ours		SEMPO	Ours
car_parts	Retail (intermittent)	12	1.207	1.130	+6.4%	1.802	1.662
hospital	Healthcare	12	0.928	0.873	+6.0%	0.088	0.081
covid_deaths	Epidemiology	30	15.579	16.389	−5.2%	0.395	0.408
nn5_daily	Banking / ATM	56	1.497	1.379	+7.9%	0.335	0.308
fred_md	Macro-economics	12	2.955	2.137	+27.7%	0.098	0.118
cif_2016	Finance / Banking	12	1.671	1.759	−5.3%	0.066	0.063
tourism_monthly	Economics / Tourism	24	3.219	2.987	+7.2%	0.296	0.284
ml_monthly	Industry / Economics	18	1.728	1.640	+5.1%	0.246	0.238
sunspot	Astronomy	30	0.099	0.089	+10.0%	57.219	51.503
saugeenday	Hydrology (river flow)	30	1.814	1.651	+9.0%	0.538	0.490
Mean			3.070	3.003	+2.2%	0.429	0.406
Median			1.700	1.646	+3.2%	0.296	0.284

A ADDITIONAL IMPLEMENTATION DETAILS AND HYPERPARAMETERS

Table 6 provides the complete hyperparameter specifications used during Stage-1 pre-training and Stage-2 MoP tuning.

B BROADER MULTI-DOMAIN ZERO-SHOT RESULTS (MONASH)

Table 7 reports the complete per-dataset zero-shot results underlying Figure 3. Both HARMONIC-MoP and Baseline SEMPO are pre-trained from scratch under identical budgets and seeds on the UTSD corpus; none of the 10 Monash datasets appear in pre-training. We report MASE (primary scale-free metric) and WAPE ($\sum |e| / \sum |y|$, equivalent to the normalized CRPS for point forecasts) at each dataset’s native horizon H . HARMONIC-MoP improves MASE on 8/10 datasets (+3.2% median, +2.2% mean) and WAPE on 8/10 datasets (+4.2% median).

Table 8: Spectral-entropy back-off $\gamma(X)$ ablation across TSLib. Stochastic Avg = {ETTm1, Weather}; Periodic Avg = {ETTh1, ETTh2, Electricity}. Lower is better; **bold** marks the best value per column.

Variant	Stochastic Avg MSE	Periodic Avg MSE	Overall MSE	Overall MAE
Baseline SEMPO	0.462	0.362	0.409	0.407
Linear $\gamma(X) = 1 - S(X)$ (Ours)	0.476	0.328	0.396	0.396
Exponential $\gamma(X) = \exp(-\alpha S(X))$	0.492	0.356	0.421	0.416
Sigmoid $\gamma(X) = \sigma(wS(X) + b)$	0.492	0.356	0.424	0.418
Threshold $\gamma(X) = \mathbb{I}(S(X) < \tau)$	0.474	0.350	0.411	0.409

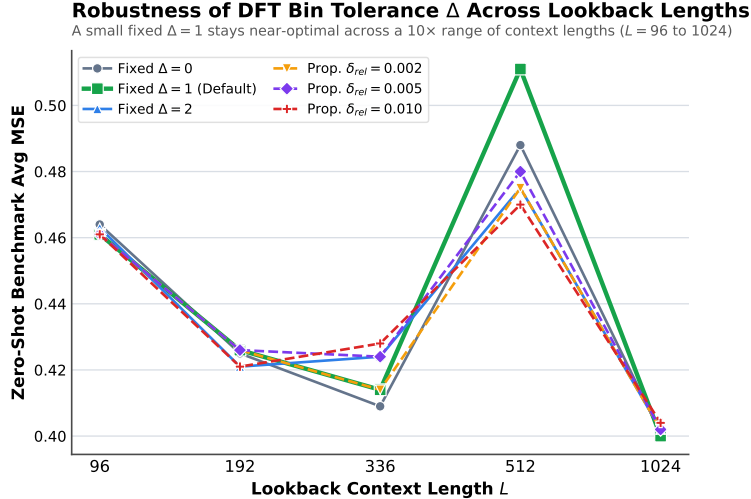


Figure 5: Zero-shot benchmark-average MSE vs. lookback length L for static ($\Delta \in \{0, 1, 2\}$) and proportional ($\delta_{rel} \in \{0.002, 0.005, 0.010\}$) DFT bin-tolerance strategies. The static default $\Delta = 1$ (green) stays near-optimal across a $10\times$ range of context lengths.

C SPECTRAL-ENTROPY BACK-OFF AND BIN-TOLERANCE SENSITIVITY

Back-off formulations. Table 8 lists the numeric values summarized in Figure 4, comparing the fixed linear default against learnable exponential, sigmoid, and dynamic-threshold parameterizations of $\gamma(X)$, all pre-trained from scratch under identical budgets and evaluated across all 7 TSLib datasets.

Bin tolerance Δ across lookback lengths. Figure 5 and Table 9 report the zero-shot benchmark-average MSE for six tolerance strategies across context lengths $L \in \{96, 192, 336, 512, 1024\}$. Small fixed tolerances remain near-optimal across the full $10\times$ range, and aggressive proportional inflation of Δ at long L slightly degrades accuracy through over-masking of adjacent non-harmonic bins.

D ADDITIONAL ABLATION FIGURES AND H-LOSS FORMULATION BREAKDOWN

Figure 6 compares the four spectral masking strategies (Ablation Plan 2), Figure 7 shows the harmonic group-count K sensitivity sweep, and Figure 8 the full 2D H-Loss weight grid (Ablation Plan 4).

Table 10 disentangles the individual terms of H-Loss under the lightweight grid-search pre-training budget (12 epochs, stride 32) referenced in Ablation Plan 4. This budget yields a lower absolute Stage-1 reconstruction MSE baseline (0.0014) than the main Plan-1 setting (0.0198).

Table 9: Zero-shot benchmark-average MSE across lookback lengths L under static and proportional bin-tolerance strategies. Parenthetical values denote the realized integer tolerance Δ . Lower is better; **bold** marks the best value per column.

Tolerance Strategy	$L = 96$	$L = 192$	$L = 336$	$L = 512$	$L = 1024$
Fixed $\Delta = 0$	0.464	0.425	0.409	0.488	0.401
Fixed $\Delta = 1$ (Default)	0.461	0.426	0.414	0.511	0.400
Fixed $\Delta = 2$	0.463	0.421	0.424	0.475	0.402
Prop. $\delta_{rel} = 0.002$	0.461	0.426	0.414	0.475	0.402
Prop. $\delta_{rel} = 0.005$	0.461	0.426	0.424	0.480	0.402
Prop. $\delta_{rel} = 0.010$	0.461	0.421	0.428	0.470	0.404

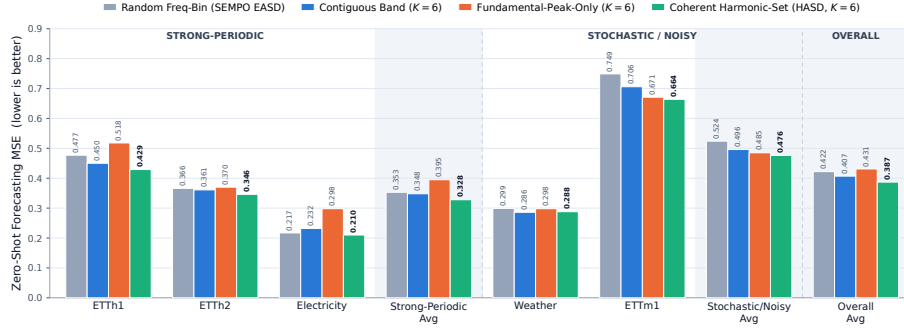


Figure 6: Downstream zero-shot MSE across four spectral masking strategies on strong-periodic (ETTh1, ETTh2, Electricity) and stochastic/noisy (Weather, ETTm1) datasets. Coherent Harmonic-Set masking (HASD, $K = 6$) consistently outperforms unstructured and peak-only baselines, achieving an overall 8.2% MSE reduction relative to the Random Frequency-Bin (SEMPO EASD) approach. Fundamental-Peak-Only masking performs *worse* than random masking (0.431 vs. 0.422), confirming that masking fundamentals without their overtones actively breaks phase lock.

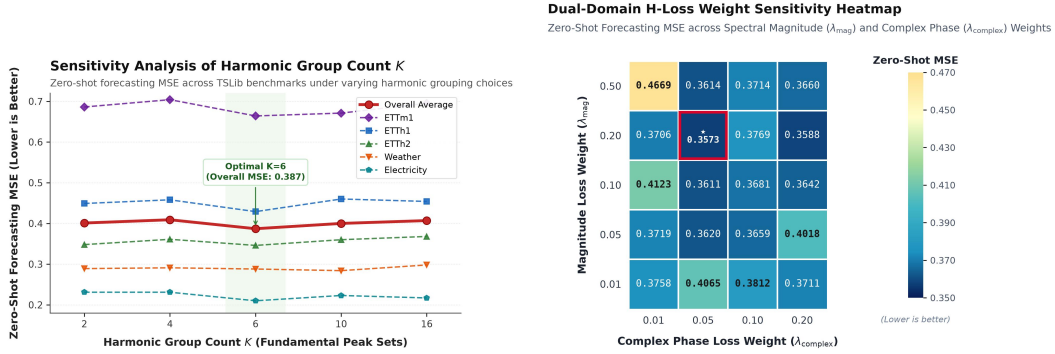


Figure 7: Sensitivity of harmonic group count K on zero-shot MSE across TSLib. The overall average is non-monotonic, optimal at $K = 6$ (0.387 MSE): too-small K under-regularizes, too-large K over-masks non-dominant frequencies.

Figure 8: H-Loss weight grid over spectral-magnitude weight λ_{mag} and complex-phase weight $\lambda_{complex}$ (average zero-shot MSE). The optimum (0.3573 MSE, red border) at $\lambda_{mag}=0.20, \lambda_{complex}=0.05$ validates our default.

E PARAMETER- AND DATA-EFFICIENCY VISUALIZATION

Figure 9 visualizes the two efficiency axes summarized in Table 3. Panel (a) plots aggregate zero-shot MSE against pre-training data scale: HARMONIC-MoP occupies the extreme data-efficient region, attaining competitive accuracy with $100\times-10,000\times$ fewer pre-training samples than TTM

Table 10: Impact of spectral loss components on Stage-1 reconstruction MSE, FFT phase MAE (radians), and downstream zero-shot forecasting error under the lightweight grid-search pre-training budget (12 epochs, stride 32). Lower is better; **bold** marks the best value per column.

Loss Formulation	Stage-1 Recon MSE	Phase MAE (rad)	ZS MSE	ZS MAE
Time-MSE Only (no H-Loss)	0.0016	0.0287	0.4214	0.4273
Magnitude-Only ($\lambda_{\text{mag}} = 0.1, \lambda_{\text{complex}} = 0$)	0.0013	0.0265	0.3952	0.4127
Complex-Only ($\lambda_{\text{mag}} = 0, \lambda_{\text{complex}} = 0.05$)	0.0017	0.0306	0.3760	0.4033
Full H-Loss L_1 ($\lambda_{\text{mag}} = 0.1, \lambda_{\text{complex}} = 0.05$)	0.0014	0.0286	0.3611	0.3943
Full H-Loss L_2 ($L_1 \rightarrow L_2$)	0.0020	0.0334	0.3695	0.4008

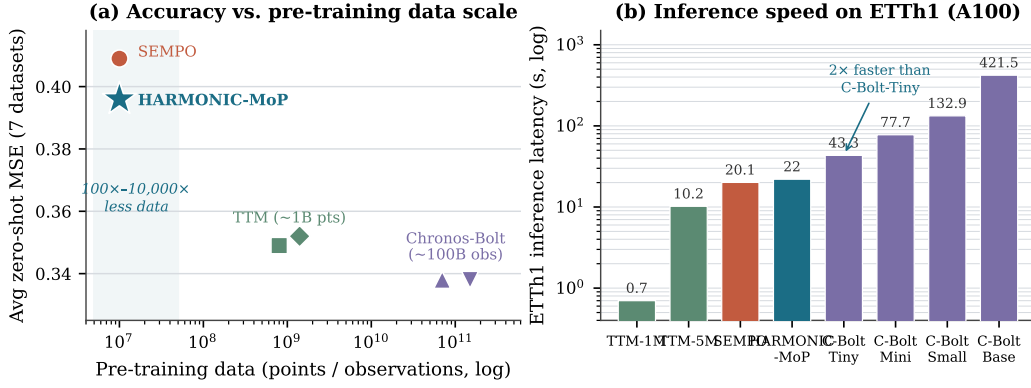


Figure 9: Efficiency of HARMONIC-MoP versus lightweight and fast foundation models. (a) Aggregate 7-dataset zero-shot MSE vs. pre-training data scale (log): HARMONIC-MoP is competitive at $100\times$ – $10,000\times$ smaller data budget. (b) ETTh1 inference latency (log): HARMONIC-MoP is $2\times$ – $19\times$ faster than the Chronos-Bolt family.

($\sim 1\text{B}$) and Chronos-Bolt ($\sim 100\text{B}$). Panel (b) shows ETTh1 inference latency on a log scale: HARMONIC-MoP is $2\times$ faster than Chronos-Bolt-Tiny and $19\times$ faster than Chronos-Bolt-Base, retaining the lightweight profile of SEMPO.

F ROBUSTNESS TO ADDITIVE OBSERVATION NOISE

This section details the additive-noise robustness study summarized in Section 4.3 (W4/Q2/S4). Additive white Gaussian noise (AWGN) has a flat power spectrum across all discrete frequency bins, which inflates normalized spectral entropy $S(X)$ toward 1 and drives the soft back-off gate $\gamma(X) = 1 - S(X)$ toward 0, potentially dampening harmonic masking on noisy periodic inputs. We inject AWGN at signal-to-noise ratios $\text{SNR} \in \{30, 20, 10, 5\}$ dB and analyze both the entropy estimator and downstream forecasting.

Zero-parameter Gaussian pre-filter. To prevent premature gate collapse, we apply a Gaussian low-pass weighting *exclusively* to the $S(X)$ -estimation path, leaving the signal fed to the Transformer backbone unchanged:

$$\tilde{E}[f] = E[f] \cdot \exp\left(-\frac{f^2}{2\sigma^2}\right), \quad \sigma = 40 \text{ bins}. \quad (12)$$

Because periodic signals concentrate energy in low-frequency harmonic bins while white noise spreads uniformly, this removes high-frequency noise power before computing $S(X)$. Table 11 shows it lifts mean gate retention at 5 dB from 56% (raw) to 87%, restoring harmonic masking on noisy periodic series (Figure 10a).

Downstream forecasting is structurally protected. HASD aggregates a clean identity spectral branch (branch weight $W \approx 0.5$) alongside the harmonic-masked branches (Eq. 5), so uncorrupted signal representations always reach the backbone regardless of $\gamma(X)$. Consequently, swapping in

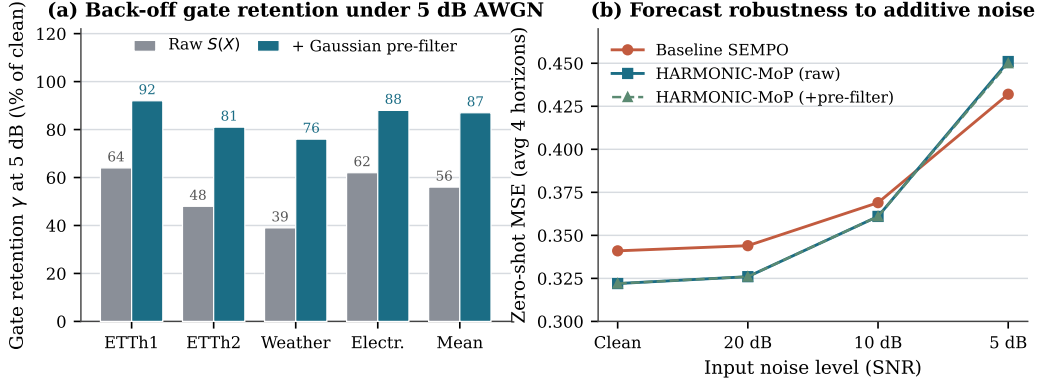


Figure 10: Robustness to additive white Gaussian noise. (a) Back-off gate retention $\gamma(X)$ at 5 dB SNR (% of clean value): the zero-parameter Gaussian pre-filter restores mean retention from 56% to 87%. (b) Zero-shot forecasting MSE vs. input SNR: HARMONIC-MoP degrades no faster than SEMPO, and the robust $\gamma(X)$ estimate leaves downstream MSE essentially unchanged.

Table 11: Back-off gate retention $\gamma(X) = 1 - S(X)$ under 5 dB AWGN (raw vs. Gaussian pre-filtered estimate). Retention is the 5 dB value as a percentage of the clean value; higher means harmonic masking remains engaged. **Bold** marks the higher retention.

Dataset	Clean γ (raw)	5 dB γ (raw)	5 dB γ (+pre-filter)	Retention (raw)	Retention (+pre-filter)
ETTh1	0.411	0.261	0.471	64%	92%
ETTh2	0.328	0.156	0.393	48%	81%
Weather	0.815	0.317	0.630	39%	76%
Electricity	0.266	0.166	0.390	62%	88%
Mean	—	—	—	56%	87%

Table 12: Zero-shot forecasting MSE (averaged over 4 horizons) across input noise levels. The robust (+pre-filter) $\gamma(X)$ estimate changes downstream MSE by ≤ 0.0005 ; HARMONIC-MoP degrades no faster than the baseline. Lower is better.

Model / Condition	Clean	20 dB	10 dB	5 dB
Baseline SEMPO	0.341	0.344	0.369	0.432
HARMONIC-MoP (raw)	0.322	0.326	0.361	0.451
HARMONIC-MoP (+pre-filter)	0.322	0.326	0.361	0.450

the robust $\gamma(X)$ estimate changes zero-shot MSE by ≤ 0.0005 across all noise levels (Table 12), and HARMONIC-MoP degrades no faster than SEMPO under increasing noise (Figure 10b). The pre-filter thus repairs the entropy estimator at zero cost, making the back-off gate fully robust for pre-training on noisy raw corpora.

G EXTENDED PER-HORIZON BENCHMARK RESULTS

Table 13 presents the full per-horizon zero-shot forecasting breakdown across all 7 TSLib datasets for forecasting horizons $H \in \{96, 192, 336, 720\}$.

Table 13: Full per-horizon zero-shot forecasting results on the TSLib benchmark comparing baseline SEMPO against HARMONIC-MoP (Ours). The published 83M SEMPO reference is included for scale context. Lower is better.

Dataset	Horizon	Baseline SEMPO		HARMONIC-MoP (Ours)		Paper Ref (83M)	
		MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.463	0.457	0.405	0.429	0.384	0.408
	192	0.480	0.469	0.424	0.440	0.409	0.426
	336	0.481	0.470	0.433	0.448	0.417	0.433
	720	0.485	0.481	0.453	0.468	0.432	0.454
ETTh2	96	0.323	0.373	0.282	0.355	0.282	0.342
	192	0.377	0.411	0.336	0.389	0.334	0.384
	336	0.393	0.426	0.364	0.411	0.355	0.403
	720	0.414	0.446	0.401	0.442	0.395	0.435
ETTm1	96	0.610	0.509	0.663	0.528	0.466	0.443
	192	0.642	0.529	0.672	0.534	0.484	0.455
	336	0.650	0.538	0.649	0.532	0.506	0.469
	720	0.654	0.548	0.671	0.552	0.557	0.498
ETTm2	96	0.229	0.312	0.230	0.319	0.196	0.286
	192	0.292	0.351	0.283	0.349	0.252	0.323
	336	0.348	0.383	0.332	0.376	0.306	0.354
	720	0.433	0.430	0.418	0.425	0.391	0.404
Weather	96	0.213	0.270	0.219	0.276	0.171	0.228
	192	0.260	0.306	0.264	0.312	0.218	0.269
	336	0.306	0.339	0.308	0.343	0.267	0.304
	720	0.359	0.373	0.361	0.377	0.336	0.350
Electricity	96	0.209	0.324	0.187	0.290	0.168	0.271
	192	0.223	0.335	0.196	0.299	0.183	0.283
	336	0.231	0.341	0.210	0.312	0.198	0.297
	720	0.261	0.359	0.248	0.341	0.238	0.329
Traffic	96	0.511	0.397	0.511	0.390	0.441	0.333
	192	0.525	0.403	0.509	0.387	0.456	0.339
	336	0.524	0.400	0.513	0.385	0.467	0.344
	720	0.546	0.404	0.543	0.395	0.503	0.360